Book - Modern digital electronics
RP JAIN

Electrical

Chapter-1

সার্কিটে রেজিস্টার না থাকলে কি হত?

⇒ বার্ন হবার সম্ভাবনা থাকে.

Inventer ***

ইনভার্টার তৈরিতে কি কি [illegible] করতে হবে?

DMOS

MMOS

Diod এর ক্যারেক্টারিস্টিক

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chapter-1

আর্কিটে রেজিস্টার না থাকলে কি হত?

⇒ বার্ন হবার সম্ভাবনা থাকে.

Inverter * * *

ইনভার্টার তৈরিতে কিকি এনালাইসিস করতে হবে?

DMOS

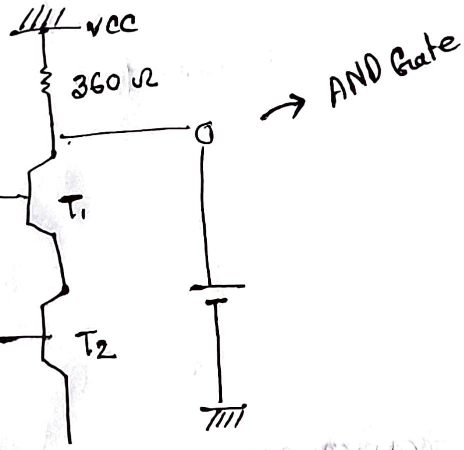
M MOS

Diod এর ক্যারেক্টারেস্টিক

Date- 03/12/2024

Digital Electronics (Tawhid sir) (2nd class)

chapter-4



A	B	ANS
0	0	1
0	1	1
1	0	1
1	1	0

মেনশন কর না ~~দি~~ দিলে 0.8 ধরে করতে হবে পরীক্ষায়।

Inverter and ~~AN~~ AND Gate ব্যবহার করে NAND Gate বানাতে হবে। (class work)

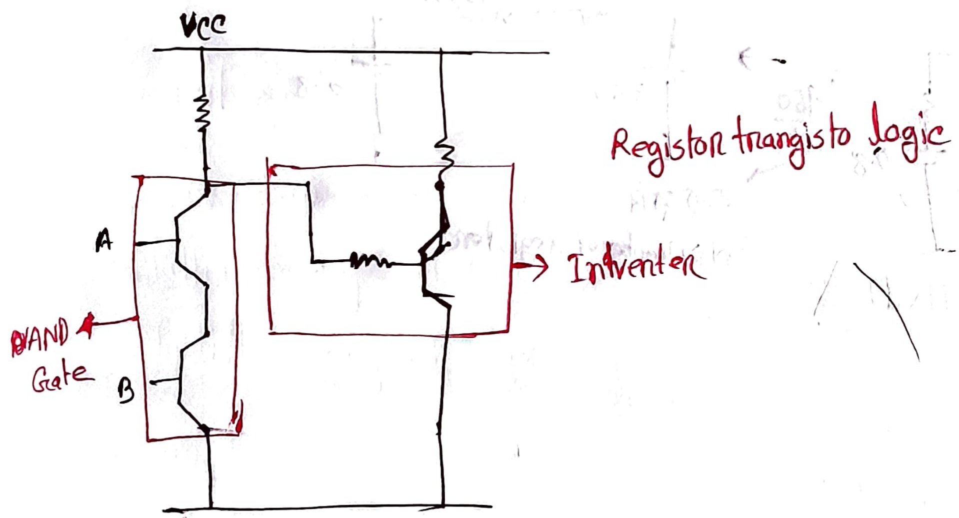


fig- NAND Gate (RTL) Regi

NAND সিরিজে যুক্ত হয়

NOR প্যারালালি যুক্ত হয়

NOR →

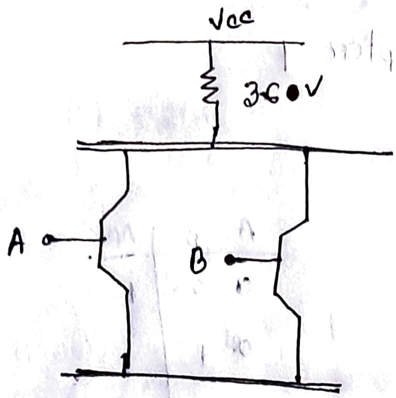
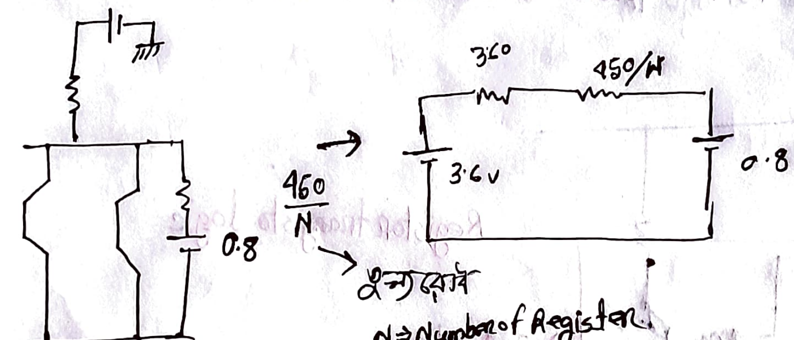


fig- NOR gate

→ OR বানাতে আরেকটি ইনভারটার যুক্ত করে দিতে হবে।

#



$\frac{460}{N}$ → ইন্সরোধ

$N \Rightarrow$ Number of Register.

$I_B = ?$

$I_C = ?$

$I_C = 9.44$ mA

$$I_B = \frac{3.6 - 0.8}{360 + \frac{450}{N}} \Rightarrow \frac{3.6 - 0.8}{360 + 90} \quad \text{when } n = 5$$

$\Rightarrow$ 6.2

$\boxed{h_{FE}} \cdot I_B \geq I_C$

→ এমপলিফিকেশন ফেক্টর।

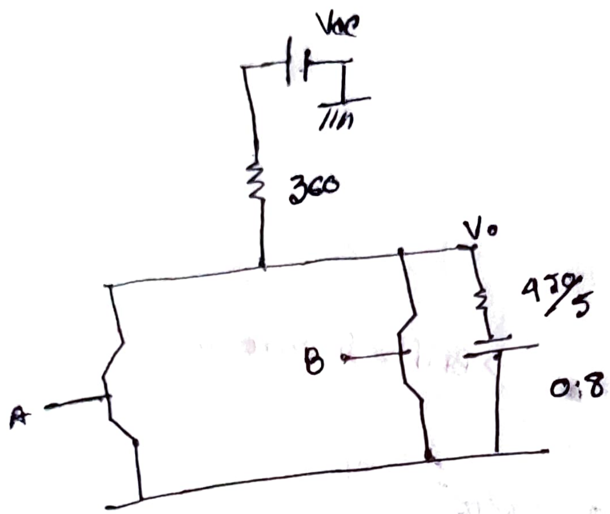
⊛ N অথবা h_{FE} যেকোন একটার মান পরীক্ষায় দেয়া থাকবে।

াকমান নম্বর আসলে পূর্বমান ধরতে হবে।

~~হবে~~

h_{FE} 6.2 ≥ 9.44

#



$I_B = \boxed{6.24 \times} \frac{450}{5} + 0.8$

মিনিমাম প্রসিয়াচে আছে A এ কারেন্ট করতে হবে.

~~কারেন্ট~~

~~$V_o = 1.36V$~~

~~I_C~~ Votage drop $= 0.8 + I_B$

$= 1.36$

Using h_{FE}, $V_o = 1.22$

difference $= (1.36 - 1.22)$

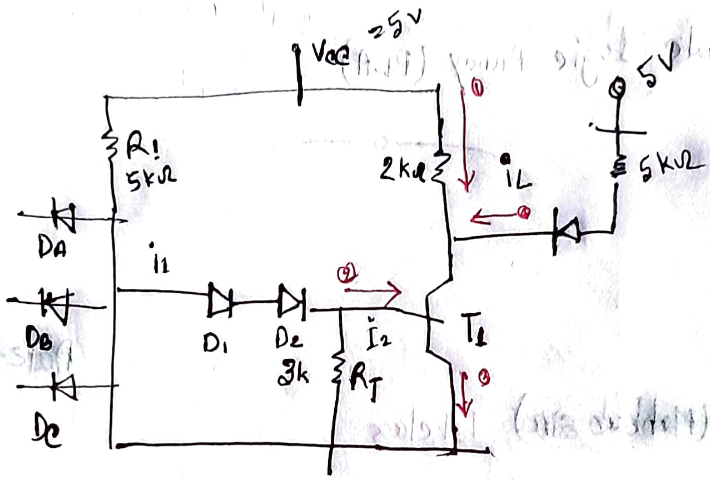
$= 0.14V$

$=$ Noise (one state)

\# সার্কিট ~~দেয়া~~ দেয়া থাকবে নয়েজ মান বের করতে হবে। ★ Exam

EEE (tawhid sir) 3rd class Chapter - 4 Date-05-012-2024

DTL দিয়ে 3 input NAND Gate →



Transistor →
BE = 0.8
BC = 0.2

$D_1 + D_2 + T_{BE} = 0.7 + 0.7 + 0.8$

$= 2.2\ V$

logically output 0 due to saturation.

cut of region
#saturation region

I_B and I_C and h_{FE} বের করতে হবে।

$-V_{cc} + i_1 R_1 + 2.2 = 0$ → কার্শফস ভোল্টেজ ল ব্যবহার করে।

$\Rightarrow i_1 = \frac{5-2.2}{5}$

$= 0.56$

$0.2 \Rightarrow T_{BC}$

$i_2 = 0.26$

$\therefore I_B = i_1 - i_2$

$= 0.3\ A$

$I_C = \frac{\cancel{5-0.2}}{2}$

$-V_{cc} + I_c R_2 + 0.2 = 0$

Then

$$I_c = \frac{5-0.2}{2}$$

$$= 2.4 \text{ A}$$

$$-V_{cc} + i_L \times 5 + \overset{\to 0.8}{0.9} = 0$$

$$i_L = \frac{5-0.9}{5}$$

$$= 0.82$$

$i_c \le h_{FE} \cdot I_B$

$i_c + N i_L \le h_{FE} \cdot I_B$ যেকোন একটার মান বের করতে হবে। বাকিগুলো কেমন থাকবে

one level noise marge

$D_1, D_2 . R_2$ cut in voltage

$= 0.6 + 0.6 + 0.5$

$= 1.7$

volage to lurance $= \overset{②}{2.7} - \overset{①}{0.9}$

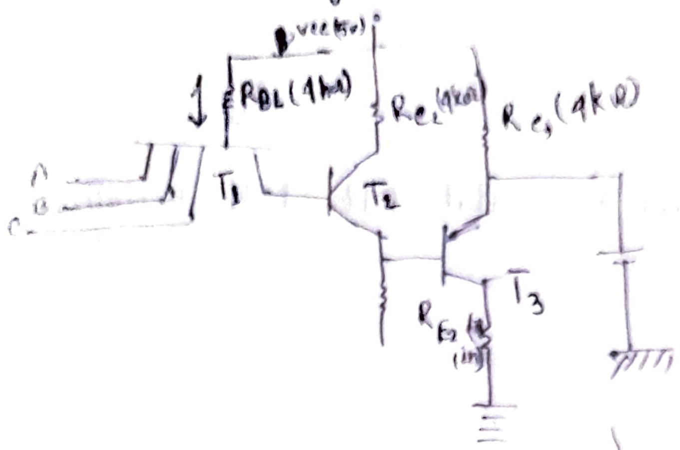
$= 0.8$

sir মান দিয়ে দিলে

Date- 09-12-2024

EEE (Tauhid sir) Class - 9

TTL (Transistor transistor logic)



Basic not draw

Working mechanism →

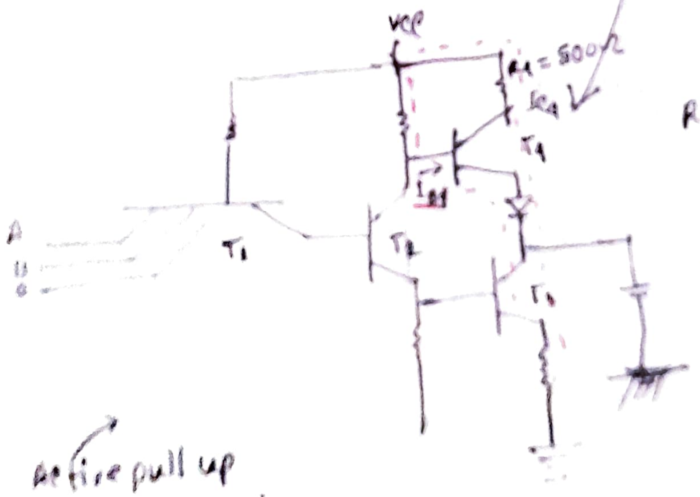
ABC তে High থাকলে consequence কি হবে?

" " low " " " "

DDL এবং TTL same working process

↳ Diod use ↳ transistor use

$\boxed{\text{RTL} \rightarrow \text{DTL} \rightarrow \text{TTL} \rightarrow \text{limitations?}}$



$R_4 = 500\Omega$

$I_{B4} = ?$

$I_{C4} = ?$

Active pull up

Totem pole

$-V_{cc} + I_{B4} R_{E2} + 0.8 + 0.7 + 0.2 = 0$

$\Rightarrow -V_{cc} + I_{B4}\, 1.4 + 0.8 + 0.7 + 0.2 = 0$

$\Rightarrow I_{B4} = \frac{5 - 1.7}{1.4} = 2.3$ mA 2.36 mA

$-5 + I_{C1} + R_1 + 0.2 + 0.7 + 0.2 = 0$

$I_{C1} = \frac{5-1.1}{500}$

$= 7.8 \text{ mA}$

chapter-4 conclusion → RTL, DTL, TTL, TTL with Active pull-up

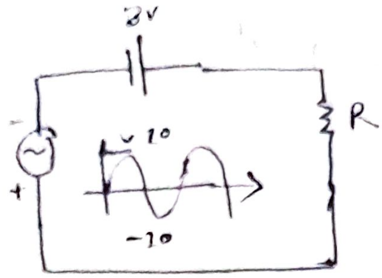
3-input AND Gate
4- input AND "
4- " OR Gate

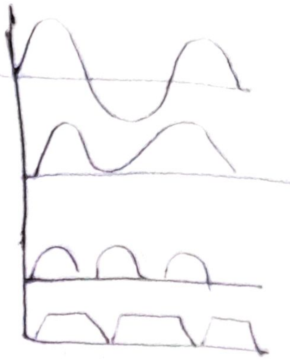
Simulation (Mahbub sir) 2nd class

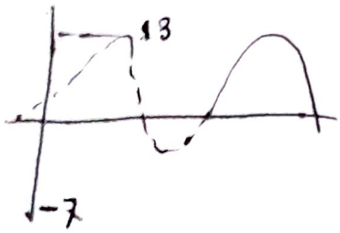
EEE (Towhid sir) (Class-5) Date-10-12-2024

Chapter-2 bojilstate

#







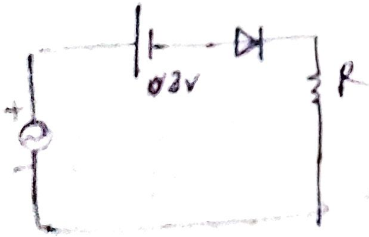
$-V_i - 3 + V_o = 0$

$V_i + 3 - V_o = 0$

$V_o = V_i + 3$

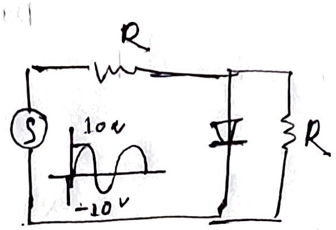
→ V_i = একবার (+) দিয়ে একবার (-) ধরে।

#

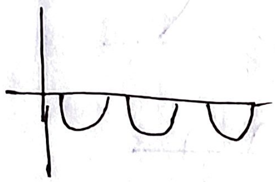


→ series cliper

#



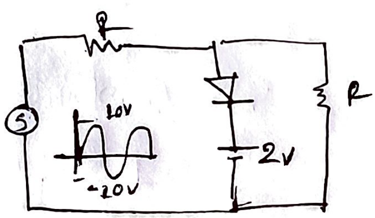
output ↓



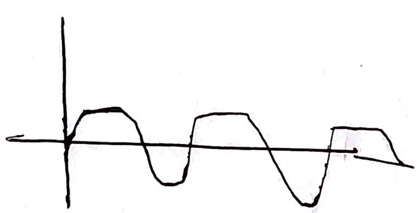
senise cliper এর উল্টো

~~forword এ পাস হয় কার্ট ক্লিপিং হয়।~~

#



Output ↓



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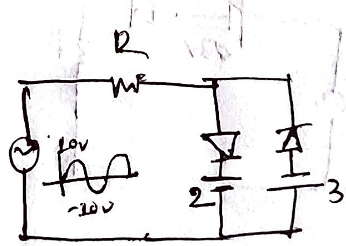
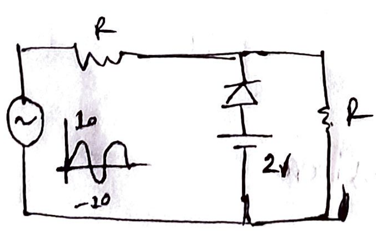
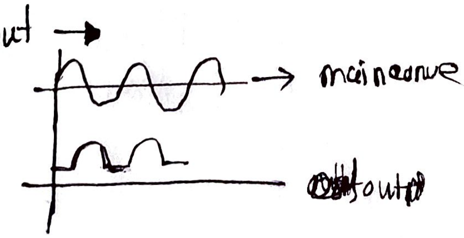


Practice

#



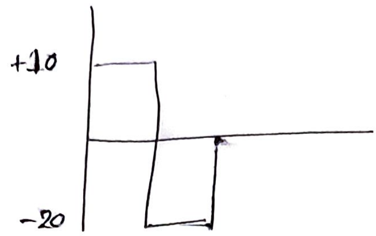
output →

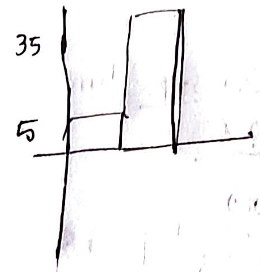


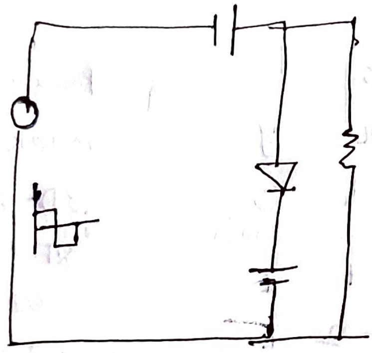
Date- 06/03/2025

EEE (Tawhid sir)

Clamper–

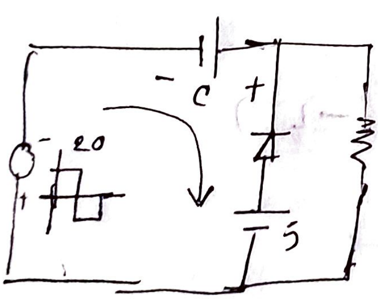






Diod forward থাকলে current pass হবে।

Diod reverse থাকলে capasitor এ voltage charge হবে

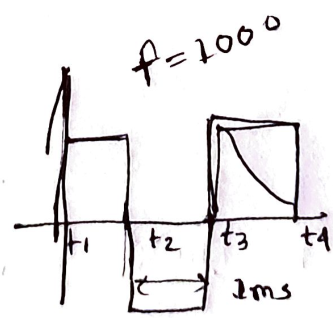


$+20 - V_c + 5 = 0$

$\Rightarrow V_c = 25\ V$

$-10 - 25 + V_o = 0$

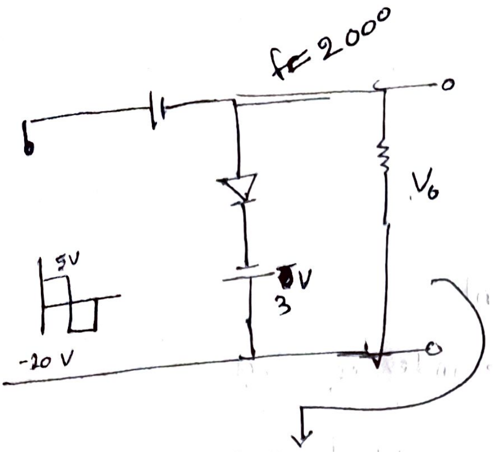
$V_o = 35$



RC হিসাব করতে হবে

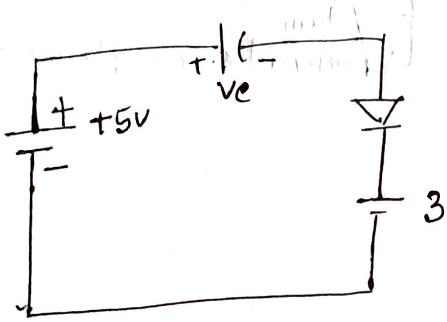
frequency দেয়া থাকতে পারে।

$T = \frac{1}{f} = 1\ ms$



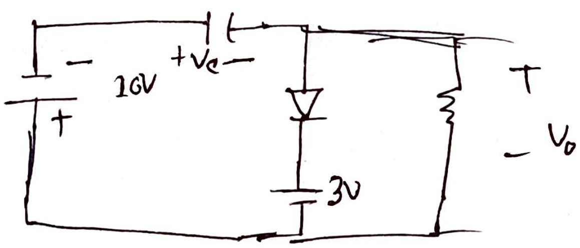
1μF

100kΩ



$-5 + V_C + 3 = 0$

$\Rightarrow V_C = 2V$



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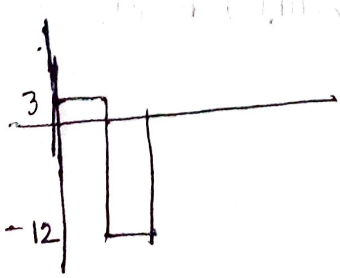
$\Rightarrow -20 + 2 + V_0 = 0$

$\Rightarrow V_0 = -12$



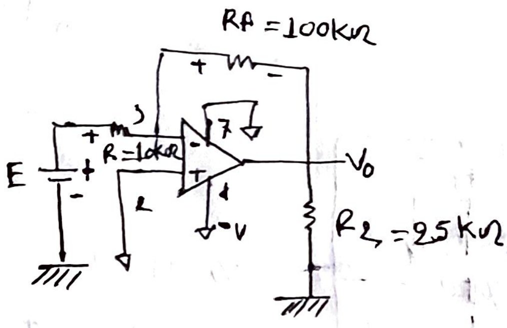
+3 থেকে -12 ব্যবধান 15

$\therefore \tau = R \times C = 1 \times 10^{-6} \times 10^{5}$

$= 10^{-1} = 0.1$

TOPIC NAME EEE (Tawhid sir) DAY Thus (09/01/2025)

TOPIC -



E = 10 V

$-E + IR_i + 0 = 0$

$\boxed{I = \frac{E}{R_i}}$

Peralel current difference = same

serise current difference = difference

$V_{Rf} = I \times R_f$

$= \boxed{\frac{E}{R_i} \times R_f}$

output

$V_0 = -V_{Rf}$

$= -\frac{E}{R_i} \times R_f$

$= \boxed{-\frac{R_f}{R_i} \times E}$

$\boxed{\frac{V_0}{R}} = \text{Gain}$ $\boxed{\therefore \frac{V_0}{R} = -\frac{R_f}{R_i}}$

I = 1

$V_0 = -100$

feedback or close loop gain

$A_{CL} \Rightarrow$ Close loop gain

$I = \frac{10}{10} = 1 \times 10^{-3} A = 1mA$

$V_0 = 100V$

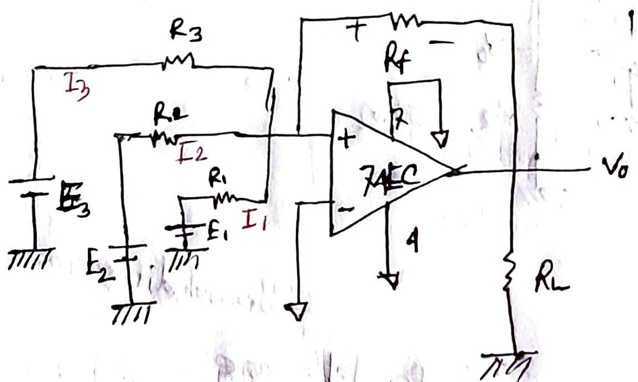
$\text{Gain} = \frac{V_0}{E} = \frac{-100}{10}$

$= -10$

$\boxed{I_0 = I + I_L \\ = I + \frac{V_0}{R_L}}$

TOPIC NAME :   DAY :
TIME :   DATE : / /

TOPIC → Inverting Adder



$I = I_1 + I_2 + I_3$

$I_1 = \frac{E_1}{R_1}$

$I_2 = \frac{E_2}{R_2}$

$I_3 = \frac{E_3}{R_3}$

$V_o = -V_{Rf}$

$= -I \cdot R_F$

$= -(I_1 + I_2 + I_3) \cdot R_f$

$= -\left(\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}\right) \cdot R_f$

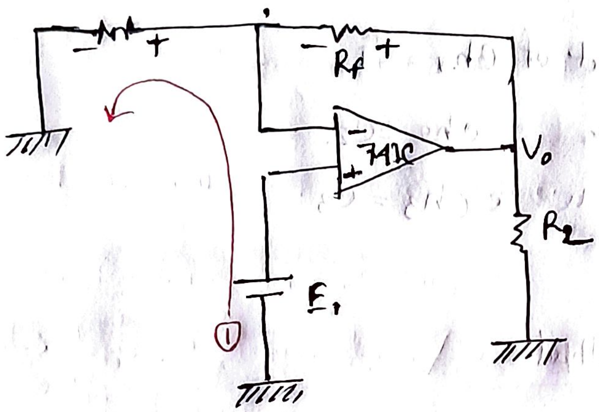
$= -(E_1 + E_2 + E_3)$ (When R_1, R_2, R_3 and R_f have same value)

chapter 1 → 1-6 page, pin 8
Chapter 2 → 1-18/19 page
Chapter 3 →
chapter 5 →

Make a substractor

TOPIC NAME : EEE (Tauhid sir) Monday (13/01/2025)
TIME : DATE :

Topic → Noninverting input



close loop gain = A_{CL}
Current flow = I
output voltage = V_0

$-E_1 + iR_1 = 0$ [Loop 1]

$I = \frac{E_1}{R_1}$

$V_{Rf} = IR_f$

$= \frac{E_1}{R_1} R_f$

$V_0 = E_1 + V_{Rf}$

$= E_1 + \frac{E_1}{R_1} \times R_f$

$= E_1 \left(1 + \frac{R_f}{R_1}\right)$

$A_{CL} = \frac{V_0}{E_1} = 1 + \frac{R_f}{R_L}$

$E_1 = 10v$

$R_f = 100 K\Omega$

$R1 = 10 K\Omega$

$R_2 = 10 K\Omega$

$V_0 = 110$

$I = 1 mA$

$A_{CL} = 11$

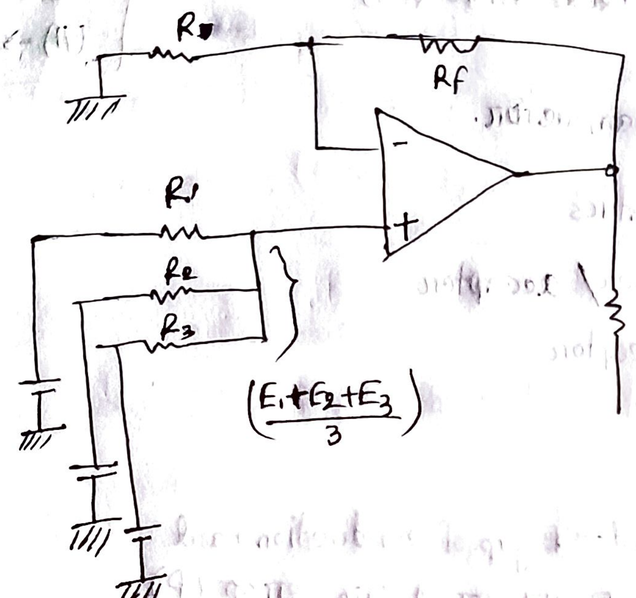
from equation

$\frac{V_0}{R_L} = I_R$ $I_R + R_L =$ output current

** Voltage flower ⇒ gain 1 হয়।

TOPIC NAME :

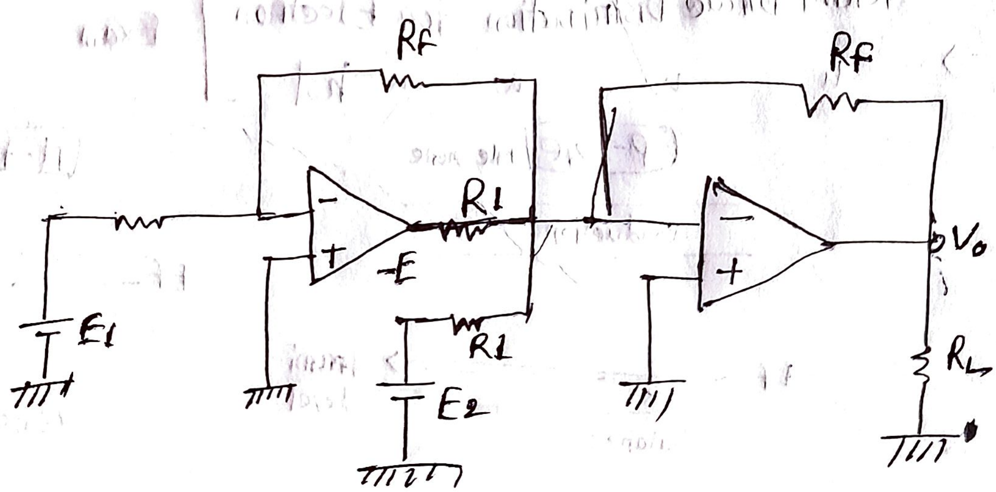
Non inverting adder



$R_F = R(n-1)$

$V_o = E_1\left(1+\frac{R_F}{R_1}\right)$

Subtractor



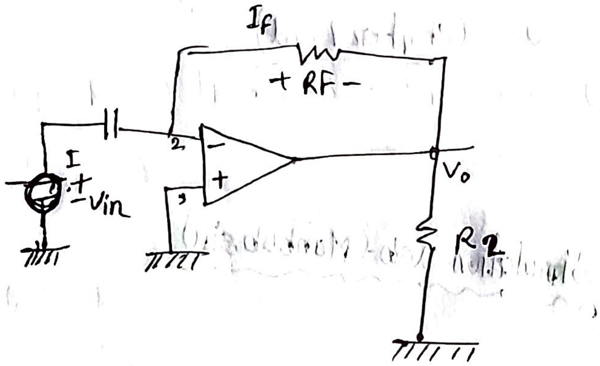
output $E_1 - E_2$

Next → differentiator and indicator

TOPIC NAME: EEE (Tawhid sir) DAY: Thus (16-01-2025)

TIME: DATE: / /

Differentiator →



$Q = CV$

$\frac{dQ}{dt} = C\frac{dv}{dt}$

$\Rightarrow I = C\frac{dv}{dt}$

$I = I_F = \frac{V_{RF}}{R_F} = \frac{-V_o}{R_F}$

$\Rightarrow C\frac{dv}{dt} = \frac{-V_o}{R_F}$

$\Rightarrow V_o = -R_F C\frac{dv}{dt}$

Problem →

(a) Design a circuit to differentiate signal from 500Hz to 1000 Hz.

$C_i = 0.1\ \mu F$.

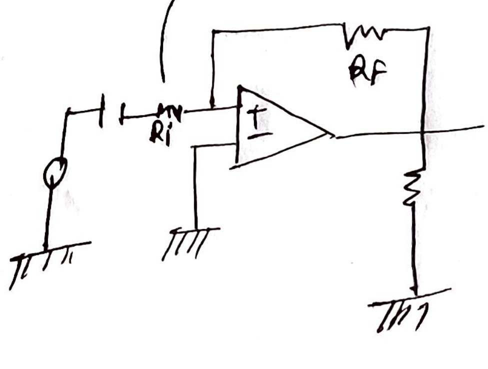
$\Rightarrow T = \frac{1}{f} = \frac{1}{500} = 0.002$ sec

$R_i = \frac{1}{2\pi f c}$

$= \frac{1}{2\pi\, 1000 \cdot 0.1\, 10^{-6}}$

$= 159.1546$

Simulation এর টাইমে Ri ব্যবহার করা হয়।



Let $T_2 = 20\, T_i$

$R_f = \frac{1}{2\pi FC}$

$= \frac{20}{2\pi \cdot 10^3 \cdot 10^{-6} \times 0.1}$

$= 31.8\, k\Omega$

$C_f = \frac{1}{2\pi f R}$

$= \frac{1}{2\pi \cdot 103 \cdot 31.8 \times 103}$

$= 0.005\, \mu F$

$\tau = RC$

$\Rightarrow \frac{1}{F} = RC$

$= \frac{1}{2\pi f} = RC$

$\Rightarrow R = \frac{1}{2\pi f C}$

$V_o = -R_f C_i \frac{dv_{in}}{dt}$ $\qquad v_{in} = 5 \sin 1000t$

$v_o = -R_f\, C_i \frac{5 \sin 1000t}{dt}$

$= 31.8 \times 0.1 \times 10^{-6}$

TOPIC NAME: EEE (Tawhid sir) DAY: Monday (20-01-2025) TIME: DATE: / /

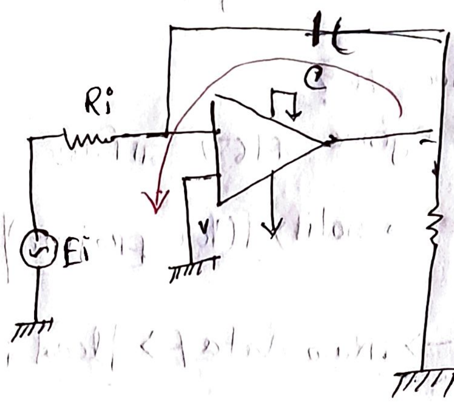
TOPIC → Integrator

$$i = \frac{E_i}{R_i}$$

$$C = \frac{Q}{V}$$

$$V_c = \frac{Q}{C}$$

$$= \frac{1}{C} \int i\,dt$$



$$V_o = -V_c$$

$$= -\frac{1}{C} \int i\,dt$$

$$= -\frac{1}{C} \int \frac{E_i}{R_i}\,dt$$

$$V_o = -\frac{1}{R_i C} \int E_i\,dt$$

Low pass filter

High pass "

Band " filter

Harmonic sinusoidal series = pulse

↳ Low pass filter

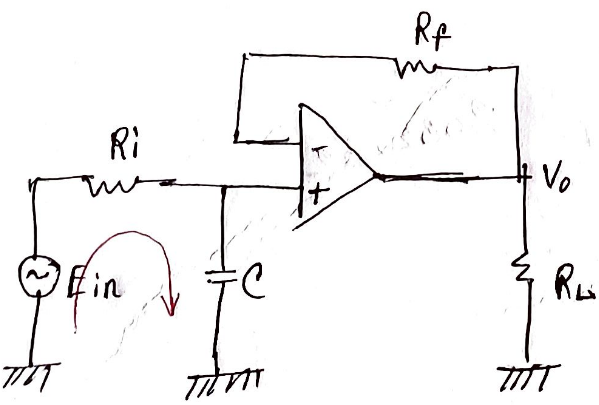


fig- lowpass filter

$$-E_i + iR_i - iJx_c = 0$$

$$i = \frac{E_i}{R_i - Jx_c}$$

$$= \frac{E_i}{R_i + \frac{1}{J\omega c}}$$

$$\boxed{\begin{aligned} X_L &= \omega L \\ X_C &= \frac{1}{\omega C} \end{aligned}} \quad J^2 = -1$$

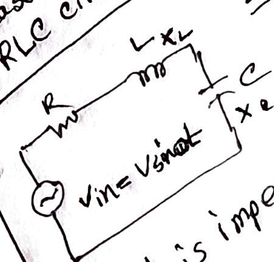
$$Z = R + J(X_L - X_C)$$

$$= \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = R + J(X_L - X_C)$$

$$= \sqrt{R^2 + (X_L - X_C)^2}$$

Reasonance RLC circuit



what is impedance

TOPIC NAME :

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$$V_c = \frac{i X_c}{R_i + \frac{1}{j\omega c}}$$

$$\frac{\frac{1}{j\omega c}}{R_i + \frac{1}{j\omega c}} \cdot E_i$$

$$\boxed{X_c = \frac{1}{2\pi f c}}$$

Peripheral device (Arpita mam)

TOPIC NAME: EEE (Tawhid sir)

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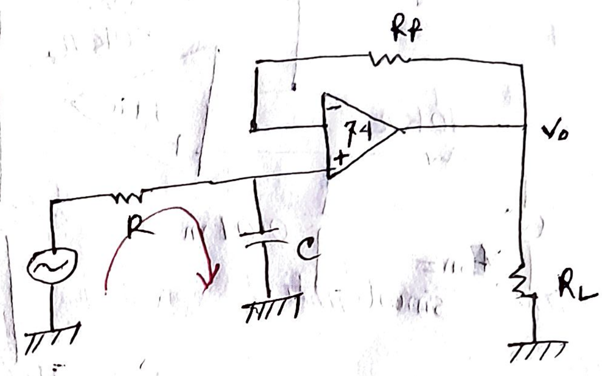
Lowpass filter

~~See priv class~~



Rc সার্কিটের impedance

$R+j(X_L-X_C)$ [Previous class]

$$Z=\sqrt{R^2+(X_L-X_C)^2}$$

$$V_C=\frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}}\times E_{in}$$

$$V_o=V_{Rf}+V_C$$

$$=0+V_C$$

$$=\frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}}\times E_{in}$$

$$=\frac{1}{1+j2\pi fRC}\times E_{in}$$

~~R = Xc = cutoff voltage = 2~~

$f_c=$ cutoff voltage

$$R=X_C=\frac{1}{2\pi f_C}$$

$$\boxed{\therefore f_c=\frac{1}{2\pi RC}}\rightarrow \text{cutoff voltage}$$

• $\log_{20}$ এর বিষয়টা বুঝি নাই।

$$V=\frac{I}{R}$$

$$I=\frac{V}{R}=\frac{V}{Z}=\frac{V}{\sqrt{R^2+X^2}}=\frac{1}{\sqrt{1^2+1^2}}=\frac{1}{\sqrt{2}}=0.7071$$

cut-off frequency - 20°

(*) Log কেন ব্যবহার করি?

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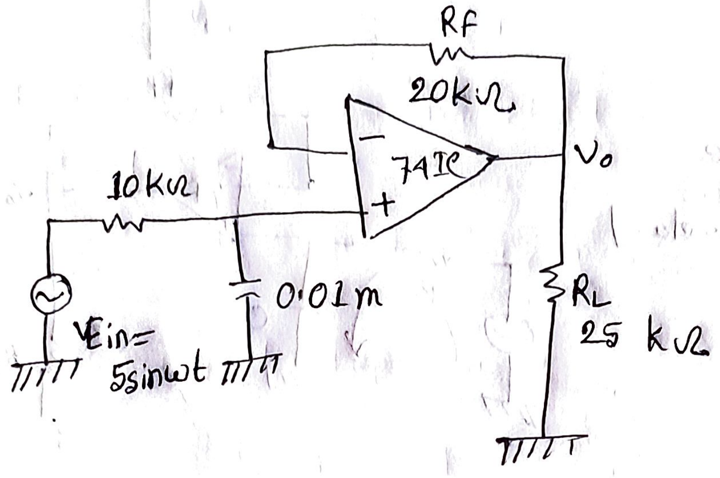
Problem solve →



$$f_c = \frac{1}{2\pi R_c} = \frac{1}{2\times\pi\times 10k\times 0.01} \quad \frac{1}{2\times 3.1416\times(10\times10^3)\times(0.01\times10^{-6})}$$

$$= 1591.545$$

$$\frac{V_o}{V_{in}} = \frac{-jx_c}{R+(-jx_c)}$$

$$= \frac{R}{\sqrt{R^2+X_c^2}}$$

$$= \frac{R}{\sqrt{R^2+R^2}}$$

$$= \frac{\cancel{R}}{\cancel{R}\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}$$

Monday 27-01-2025

clamper → signal এর লেভেল চেইঞ্জ করে।

Low pass filter → Low frequency তে কাজ করে।

↳ Highpass filter

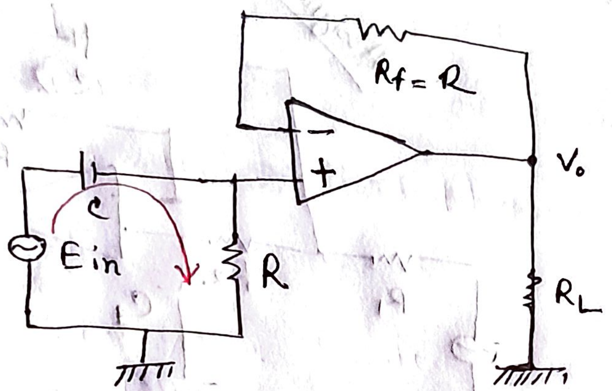
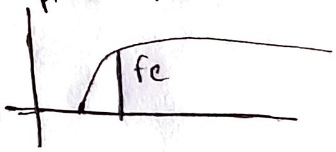


fig- highpass filter

* ২টি জাংশন Reverse ⇒ cut off
* " " forward ⇒ saturated

transistor → trans-Registor

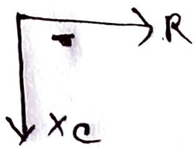


$$X_C = \frac{1}{2\pi f c}$$

Appling KVL →

$$-E_{in} - jX_C I + I \cdot R = 0$$

$R - jX_C$



$$\therefore I = \frac{E_{in}}{R - jX_C}$$

$$= \frac{E_{in}}{R - j\frac{1}{\omega C}}$$

$$V_{in} = I \cdot R$$

$$= \frac{E_{in}}{R - j\frac{1}{\omega C}} \cdot R$$

$$= R \frac{E_{in}}{1 - \frac{j}{\omega R C}}$$

$$V_o = \frac{E_{in}}{1 - \frac{j}{\omega R C}}$$

$$\therefore A_{CL} = \left|\frac{V_o}{E_i}\right| = \frac{V_o}{E_{in}} = \frac{1}{1 - \frac{j}{\omega R C}}$$

Cutofof voltage →

$R = X_C$

$R = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

$f_c = \frac{1}{2\pi R C}$

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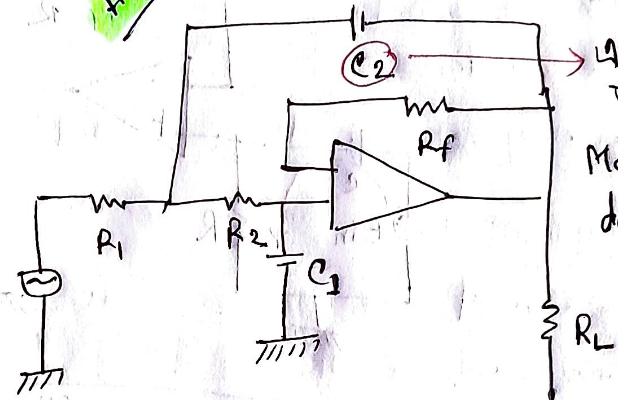
ক্যাপাসিটর টা কেন দেয়া হয়েছে?

ACL কিভাবে আসলো? Details

Assignment



এটা কেন দিছে? জানতে হবে। Mathmatically derivation সহ।

fig— Butterworth filter (lowpass)

TOPIC NAME : EEE (Tawhid sir)   DAY : Tue (30-01-2025)

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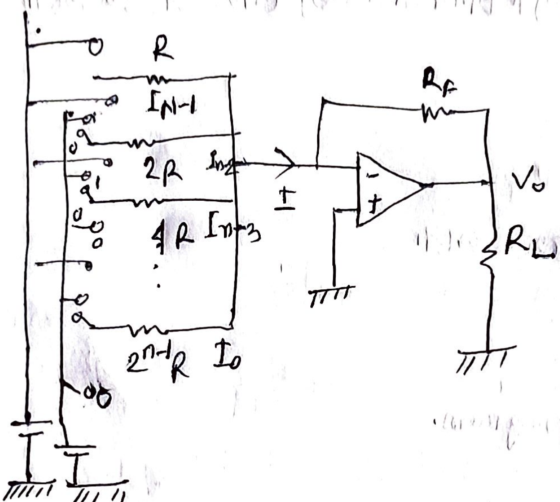
TOPIC - Weighted Register D/A converter.   RTL বইয়ের chapter 10



V 0100

$I = I_3 + I_2 + I_1 - I_0$

$V_0 = \frac{-R_F}{R} V_{in}$ [Inverting এর মতো]

$\Rightarrow$

$V_0 = -(-V_R)\left[\frac{R_F}{R} b_{n-1} + \frac{R_F}{2R} b_{n-2} + \cdots \frac{R_F}{2^{n-1}R} b_0\right]$

$= -(-V_R)\frac{R_F}{2^{n-1}R}\left(2^{n-1} b_{n-1} + 2^{n-2} b_{n-2} + \cdots b_0\right)$

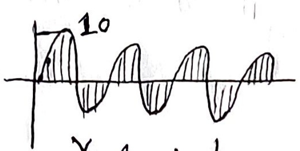
$I_3 = \frac{V_3}{R}$

$\vdots$

$I_0 = \frac{V_0}{2^{n-1}R}$

$\therefore V_0 = -|-V_R|\left(\frac{R_F}{R}\cdot b_{n-1} + \frac{R_F}{2R} b_{n-2} + \cdots \frac{R_F}{2^{n-1}R} b_0\right)$

TOPIC NAME : ~~Taw~~ EEE (Tawhid sir)　　DAY :
TIME :　　DATE : / /



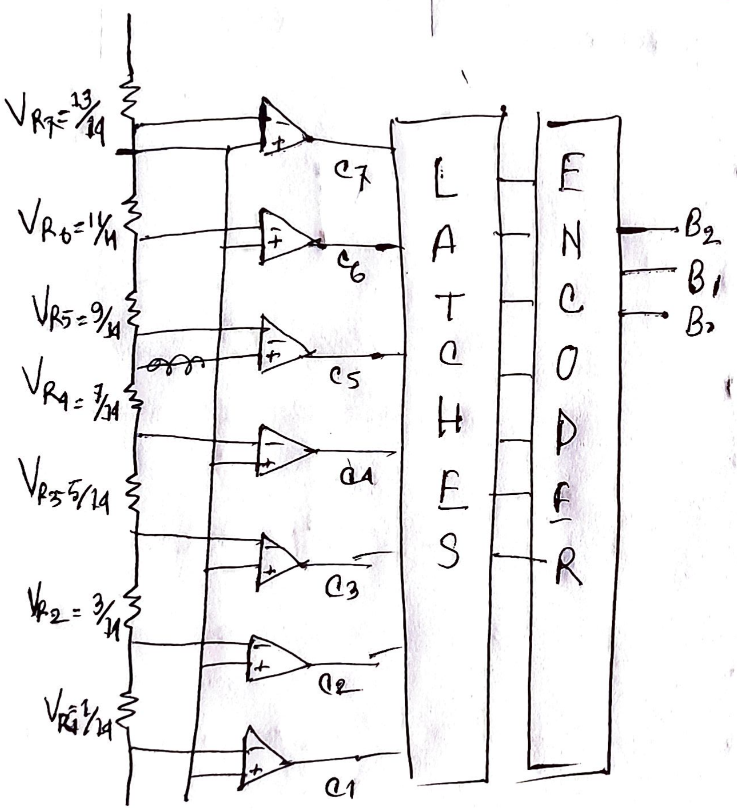
$Y = 10 \sin \omega t$

[Sampling]

Course Done today

1. Sampling
2. Quentization
3. Encoding

class not clear
TOME



Quentization এ Error কমানোর উপায় কি? Exam